

HeliHero Reflection

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[GitHub Link](#)

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Motivation and Background

HeliHero was inspired first by arcade-style lander games, like *Jetpack Joyride* and *ChopLifter*, and even *Hill Climb Racing*. The goal was to modernize the game without losing the retro-feel. Many popular games today succeed by combining simple mechanics and retro-style visuals with the smooth physics and clarity enabled by modern technology. We wanted to create a game that was visually appealing, simple to understand, but fun, challenging, and free flowing. Games where you have freedom of movement always feel more life-like. The mix of retro-style artwork and simple imagery with modern movement flow created a fun, exciting game for people of all ages.

In our physics-based lander game, players control a military helicopter through a war zone, being chased by missiles. The player's goal is to navigate the 2D scrolling world and safely reach the landing pad. Users have a limited amount of fuel, however you can pick up fuel sprites along the way, resetting your fuel bar. Similarly, coin sprites are dispersed throughout the map, collecting them will improve your score. The user must land the helicopter at a slow speed, and level, otherwise it is a crash. Colliding with an incoming missile, or the ground, will also result in a crash. Users use the left and right arrow keys to rotate the helicopter forward and back, and the up arrow key as a thruster, propelling the helicopter in a given direction. Missiles are fired at the helicopter from the ground, and will track the helicopter in the air. Once the user lands on the landingPad, they are taken to the next level. Each level has harder terrain, more missiles, and less fuel sprites. Culminating in the final level, which includes "decoy fuels," meaning that if you take too many wrong turns, you won't have enough fuel to finish.

Zach McCown will oversee timeline, manage Github repo, and contribute to artwork and helicopter interactions. Will Bruzzese will create art, and assist with programming. Weston Hollmann will assist with artwork, and contribute to level logic. Given that this is a class assignment and the goal is to learn as well as create a polished game, each team member will assist with the other's tasks when needed, to improve design flow and learn from one another.

Game Development

For our design process, our goal was to complete one level almost entirely, then use that format for our next levels, changing terrain, missiles, and fuel sprites/coins to increase difficulty. It is important to note that while we each were responsible for certain sections to commit (sections will generally be described below), we worked together throughout the entire process, doing the vast majority of the work in a room together, collaborating on code and sharing ideas. Each group member contributed to every section, and contributions were very equally balanced.

We started by adding a background and helicopter with realistic physics movement. Zach and Weston handled most of the early documentation and Helicopter movement. In our first demo we got consistent feedback that the realistic physics of the helicopter was very cool, but too sensitive. We later toned this down. Will added fuel sprites, coins, scoring logic, and the system for detecting and handling crash conditions during landing and collisions. This provided our game with more functionality, but very little difficulty. Zach and Weston continued to improve the UI after this, we knew we needed to add difficulty at some point, but we had a plan to do that with missiles. Knowing this, these two tackled more complicated issues such as explosion visuals, fuel bar and scoreboard, as well as an ending screen.

After this, our level 1 was fairly polished, we just needed to add some difficulty. We did this by adding missiles. Will added the sprites and initial logic for both the missile silo and missile itself, with Weston and Zach each following up to add some of the visuals and game-ending logic. Finally, it was time to add more levels. This included new backgrounds (Generated by Will and Zach), new terrain, and newly placed fuel/coin sprites. Zach also had the idea on the 3rd level to add a new dynamic, fake fuel sprite. Which don't actually recharge your fuel bar. This added a new layer of difficulty and thinking, to keep the user from feeling as though it was too repetitive. Weston added final details like a homescreen and back/home buttons, and with that, our game was complete.

Final Gameplay

[Youtube Link](#)

Future Work

We all agreed that we thoroughly enjoyed developing this game. It was so fulfilling to watch it come together, and to be proud of what you created. We noted that it was really cool to know exactly what each level entailed, exactly what went into making it, and STILL enjoying playing it. We definitely faced some growing pains with Unity, just from trying new features and learning how powerful it actually is. As well as getting the game to work on everyone's machine. We solved both of these with a lot of failure and collaborative learning. We had a great work environment, where we weren't afraid to mess up or try something new. A lot of moments where a group member would chime in "I just figured out how to do this last night," or "check out this new feature I learned on Unity that we could use to add..." Our biggest strength was our collaboration, and that is how we overcame our challenges.

Initial comments on our game were primarily divided into three sentiments: love the idea, need more challenges, and helicopter is too sensitive. The first one was important, we knew we were on the right track. Once we refined helicopter sensitivity, our next major design question was how to maintain engagement and escalate challenges. We did this first through lowering fuel level, then through missiles, and finally through puzzling challenges like fake fuel sprites. We found that this created a very good balance of anxiety for the user and sense of accomplishment, keeping them playing longer.

If we could continue working on our game, we would add more forms of challenge, such as "dumb" vs "smart" missiles. Some that chase you, and some that just shoot in a single direction. We would also add more levels, with better graphics, and more interactivity. This would include keys that you need to collect to open gates, tunnels where you can't fly too high or too low, and passengers to carry to safety. Turning it into less of a pure linear game, and more of a real-life interactive feel, without losing the timeline, 2d-sliding narrative that we built the game on.

Overall, we fully enjoyed the process of planning, building, testing, and refining the game. It's a great feeling to build something that others enjoy playing, and we look forward to having more opportunities to refine HeliHero in the future.